

PHYS 143 – Mechanics and Thermal Physics

Vectors

How do we describe, combine and calculate with quantities that have both magnitude and direction?

Hope Donglo

Office 24, Department of Physics, University of Ghana

Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 1, §1.7–1.10

- **Lecturer:** Hope Donglo, Office 24, Department of Physics.
- **Text:** Young and Freedman, *University Physics*. Copies in the Physics Department Library and Balme Library.
- **Sakai** (sakai.ug.edu.gh): slides, problem sheets, announcements, office hours. Log in this week.
- **Attendance** is required; ten absences without a UG physician's note bars you from the final exam.
- **Homework** must be your own work. Asking lecturers, TAs and classmates for help is encouraged.

Everything else

Venues, times, TA names and the assessment breakdown are on Sakai.

By the end of this week, you should be able to:

1. Represent, add and subtract vectors graphically.
2. Resolve vectors into components and reconstruct their magnitude and direction.
3. Add vectors using components and unit-vector notation.
4. Calculate and interpret dot and cross products.

Throughout the week, use

**IDENTIFY → SET UP → EXECUTE →
EVALUATE**

and for each new idea: QUESTION → SKETCH →
PREDICT → CALCULATE → EVALUATE

Three contacts:

1 UNDERSTAND

Why do we need vectors?

2 CALCULATE

How do we calculate with vectors accurately?

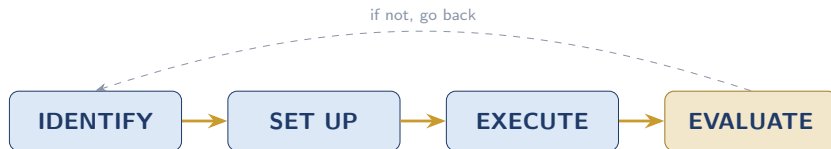
3 EXTEND

How do vector products describe physical quantities?

Contact 1: Understand

Why do we need vectors?

How we solve every problem this semester



IDENTIFY

What is being asked?
What quantities are known?
What physics is relevant?

SET UP

Draw a diagram.
Choose coordinates.
Predict the approximate size and direction.
Select the equations.

EXECUTE

Perform the calculation.
Carry units.
Keep track of signs.

EVALUATE

Are the units correct?
Is the magnitude reasonable?
Does the direction agree with the sketch?
Does it make physical sense?

This is Young and Freedman's problem-solving strategy. Every substantial worked example this semester follows these four steps, and so should your written solutions.

An opening problem

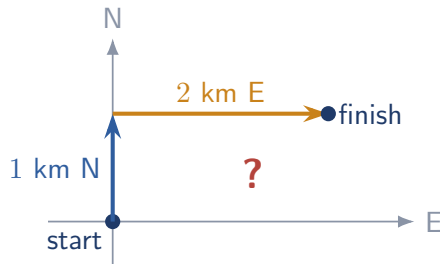
Opening problem

You walk **1 km north**, then **2 km east**.

Where are you now, relative to where you started?

Predict first

- Are you 3 km from the starting point?
- Is “2.24 km” by itself a complete answer?
- In approximately what direction should the answer point?



We return to this after graphical vector addition.

Scalar

A scalar quantity is **described completely by a numerical value and appropriate units**, where applicable.

e.g. mass, time, temperature, distance, speed, energy.

Notation

$\vec{A}$ = the vector

$A = |\vec{A}|$ = magnitude of the vector

Print often uses bold A for $\vec{A}$.

Vector

A vector quantity **requires both magnitude and direction**.

e.g. displacement, velocity, acceleration, force.

Keep these distinct

The magnitude of a vector is a **scalar** and is **never negative**.

Concept check: scalar or vector?

Decide for each pair, then compare with your neighbour

- | | |
|---------------------------|-------------------------------|
| (a) 15 m/s | (b) 15 m/s north |
| (c) distance walked, 3 km | (d) displacement, 3 km east |
| (e) temperature, 30°C | (f) a force of 20 N downwards |
| (g) -5 J of energy | (h) mass, 70 kg |

Concept check: scalar or vector?

Decide for each pair, then compare with your neighbour

- | | |
|---------------------------|-------------------------------|
| (a) 15 m/s | (b) 15 m/s north |
| (c) distance walked, 3 km | (d) displacement, 3 km east |
| (e) temperature, 30°C | (f) a force of 20 N downwards |
| (g) -5 J of energy | (h) mass, 70 kg |

Answers

Scalars: (a) speed, (c) distance, (e), (g), (h). Vectors: (b) velocity, (d) displacement, (f) force.

A negative number is not automatically a vector: -5 J is still a scalar.

Distance and displacement

Distance

scalar
depends on the
total path travelled

Displacement

vector
runs directly from
initial position to
final position

Question

You walk once around a 400 m track and stop where you started. What is your displacement?

Distance and displacement

Distance

scalar
depends on the
total path travelled

Displacement

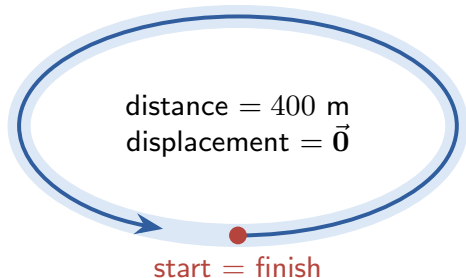
vector
runs directly from
initial position to
final position

Question

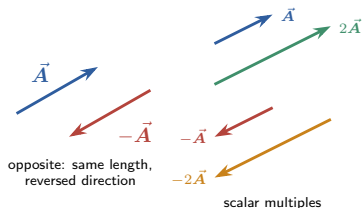
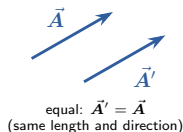
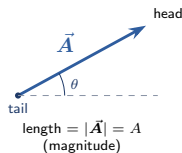
You walk once around a 400 m track and stop where you started. What is your displacement?

Physical meaning

Zero, although the distance travelled is 400 m.



Representing a vector



Rule: equal and opposite vectors

Two vectors are **equal** if they have the same magnitude and the same direction, wherever they are drawn.

$-\vec{A}$ has the same magnitude as $\vec{A}$ and the opposite direction.

Rule: multiplying by a scalar

$c\vec{A}$ has magnitude $|c|A$.

$c > 0$: the direction is unchanged ($2\vec{A}$).

$c < 0$: the direction is reversed ($-\vec{A}$, $-2\vec{A}$).

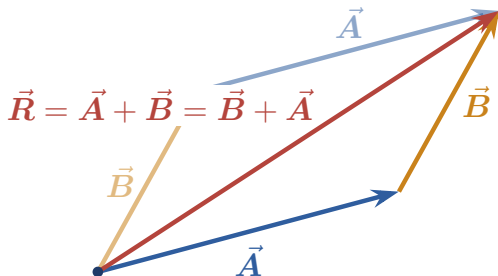
This prepares us for expressions such as $2\vec{D} - \vec{E}$.

Adding vectors graphically: head to tail

Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?

1. Draw the first vector.
2. Place the tail of the next vector at the head of the first.
3. Continue for all vectors.
4. Draw the resultant from the starting point to the final point.

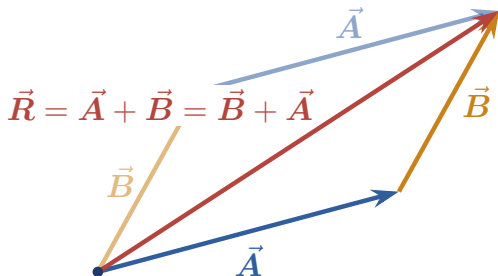


Adding vectors graphically: head to tail

Predict first

If $\vec{A}$ and $\vec{B}$ are swapped, does the resultant change?

1. Draw the first vector.
2. Place the tail of the next vector at the head of the first.
3. Continue for all vectors.
4. Draw the resultant from the starting point to the final point.



Commutative law

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \text{ (the parallelogram)}$$

Magnitudes do not simply add

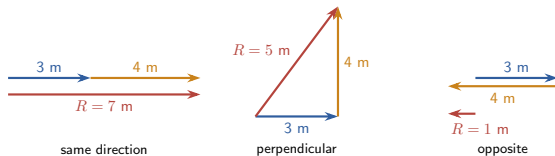
Predict first

Two vectors have magnitudes 3 m and 4 m. Must their resultant be 7 m?

Magnitudes do not simply add

Predict first

Two vectors have magnitudes 3 m and 4 m. Must their resultant be 7 m?

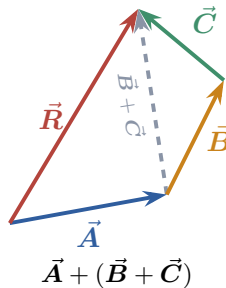
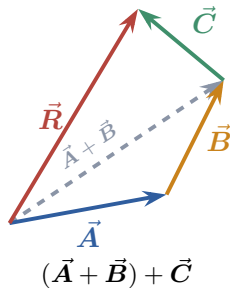


No. In general

$$|\vec{A} + \vec{B}| \neq |\vec{A}| + |\vec{B}|$$

The magnitudes add directly only when the vectors point in the same direction. For 3 m and 4 m, $1 \text{ m} \leq |\vec{A} + \vec{B}| \leq 7 \text{ m}$ (Toolkit 2).

Three or more vectors: associativity



Associative law

$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$. The vectors may be grouped in any way; the resultant is the same.

Subtracting vectors

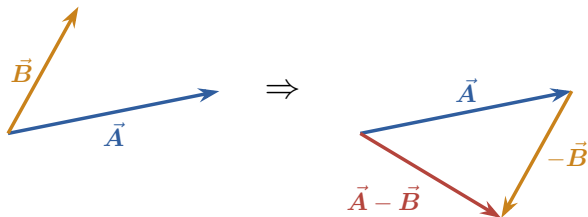
Rule: subtraction

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Reverse $\vec{B}$, then add head to tail.

Physical meaning

$\vec{A} - \vec{B}$ is the vector that takes you from the head of $\vec{B}$ to the head of $\vec{A}$ (when they share a tail). This is how *change in velocity* $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ will be found.



Returning to the opening problem

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Returning to the opening problem

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Execute:

$$R = \sqrt{(1.00 \text{ km})^2 + (2.00 \text{ km})^2} = 2.24 \text{ km}$$

$$\tan \theta = \frac{2.00 \text{ km}}{1.00 \text{ km}} \Rightarrow \theta = 63.4^\circ \text{ east of north}$$

Returning to the opening problem

Identify: asked for the displacement $\vec{R}$ (magnitude *and* direction); known: two perpendicular legs.

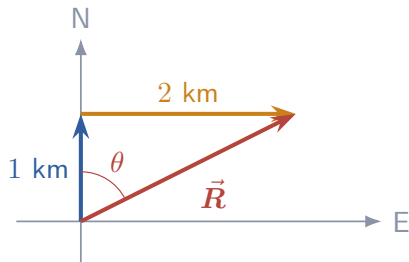
Set up: sketch head to tail, axes N and E. Predict: $2 < R < 3$ km, closer to east. Pythagoras and $\tan$.

Execute:

$$R = \sqrt{(1.00 \text{ km})^2 + (2.00 \text{ km})^2} = 2.24 \text{ km}$$

$$\tan \theta = \frac{2.00 \text{ km}}{1.00 \text{ km}} \Rightarrow \theta = 63.4^\circ \text{ east of north}$$

Evaluate: units km; $2 < R < 3$ km as predicted; direction closer to east, matching the sketch.



Young, Example 1.5

TOOLKIT 1 – Directions and angles

$$E = 0^\circ, \quad N = 90^\circ, \quad W = 180^\circ, \quad S = 270^\circ$$

- “30° north of east” → start east, turn north.
- “30° east of north” → start north, turn east.
- Standard θ : measured counter-clockwise from $+x$.
- Always sketch before converting.

Predict first

Convert to θ : 30° east of north, 40° south of west, 20° west of south.

TOOLKIT 1 – Directions and angles

$$E = 0^\circ, \quad N = 90^\circ, \quad W = 180^\circ, \quad S = 270^\circ$$

- “30° north of east” $\rightarrow$ start east, turn north.
- “30° east of north” $\rightarrow$ start north, turn east.
- Standard θ : measured counter-clockwise from $+x$.
- Always sketch before converting.

Predict first

Convert to θ : 30° east of north, 40° south of west, 20° west of south.

$$\theta = 60^\circ, \quad \theta = 220^\circ, \quad \theta = 250^\circ.$$

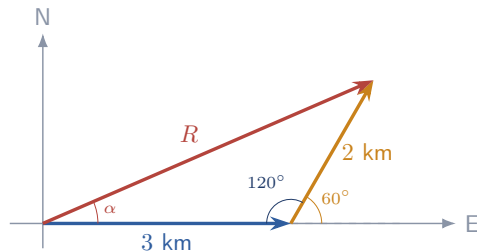
Your turn: when the angle is not 90°

Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

Predict first

Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?



Your turn: when the angle is not 90°

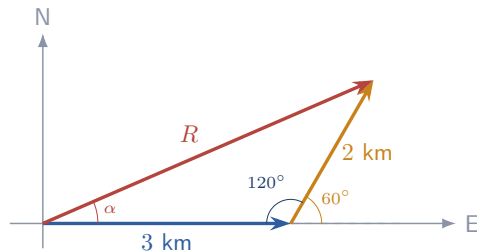
Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

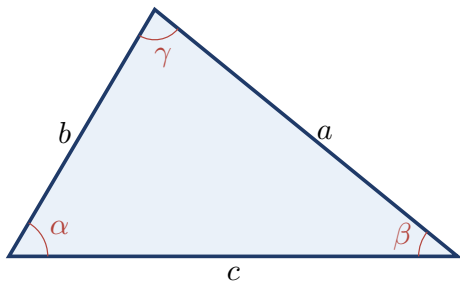
Predict first

Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?

The triangle has no right angle: the angle *inside* the triangle between the two legs is $180^\circ - 60^\circ = 120^\circ$, not the 60° given. Pythagoras does not apply directly; we need a tool for general triangles.



A tool for vector triangles: the cosine and sine rules



side a is opposite angle α , etc.

Cosine rule: find a side

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

Sine rule: find an angle

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

These are mathematical tools, not a new vector idea; we use them without derivation. If $\alpha = 90^\circ$ the cosine rule becomes Pythagoras.

TOOLKIT 2 – Vector triangles

1. Construct. Draw the vectors head to tail; the resultant runs from the first tail to the last head.

2. Find the internal angle of the triangle.

3. Choose the tool.

Right triangle: Pythagoras and basic trigonometry.

General triangle: cosine rule for a side, sine rule for an angle.

WATCH OUT: always identify the actual internal triangle angle before using the cosine or sine rule.

4. Sanity check. For the magnitude of the sum of two vectors:

$$|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$$

For 3 m and 4 m: $1 \text{ m} \leq |\vec{A} + \vec{B}| \leq 7 \text{ m}$, so a calculated 9 m must be wrong.

Predict first

In the cyclist problem the given angle is 60° . What is the internal angle opposite R ?

TOOLKIT 2 – Vector triangles

1. Construct. Draw the vectors head to tail; the resultant runs from the first tail to the last head.

2. Find the internal angle of the triangle.

3. Choose the tool.

Right triangle: Pythagoras and basic trigonometry.

General triangle: cosine rule for a side, sine rule for an angle.

WATCH OUT: always identify the actual internal triangle angle before using the cosine or sine rule.

4. Sanity check. For the magnitude of the sum of two vectors:

$$|A - B| \leq |\vec{A} + \vec{B}| \leq A + B$$

For 3 m and 4 m: $1 \text{ m} \leq |\vec{A} + \vec{B}| \leq 7 \text{ m}$, so a calculated 9 m must be wrong.

Predict first

In the cyclist problem the given angle is 60° . What is the internal angle opposite R ?

$180^\circ - 60^\circ = 120^\circ$, which is why the cosine rule uses $\cos 120^\circ$.

Your turn: solution

Set up: sketch; the 120° angle is opposite R . Predict $3 < R < 5$ km.

Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00) \cos 120^\circ = 19.0$$

$$R = 4.36 \text{ km}$$

Your turn: solution

Set up: sketch; the 120° angle is opposite R . Predict $3 < R < 5$ km.

Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00) \cos 120^\circ = 19.0$$

$$R = 4.36 \text{ km}$$

sine rule for the direction α at the start:

$$\frac{\sin \alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \Rightarrow \alpha = 23.4^\circ$$

$\vec{R} = 4.36 \text{ km}$ at 23.4° north of east.

Your turn: solution

Set up: sketch; the 120° angle is opposite R . Predict $3 < R < 5$ km.

Execute, cosine rule:

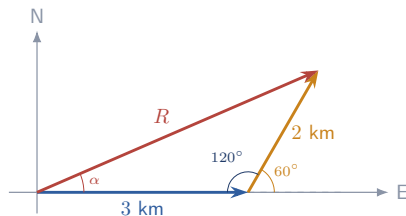
$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00) \cos 120^\circ = 19.0$$

$$R = 4.36 \text{ km}$$

sine rule for the direction α at the start:

$$\frac{\sin \alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \Rightarrow \alpha = 23.4^\circ$$

$\vec{R} = 4.36$ km at 23.4° north of east. **Evaluate:** units km; $3 < R < 5$ as predicted; $0^\circ < \alpha < 60^\circ$, matching the sketch. ✓



Now suppose we have four vectors

Question

A delivery rider makes four legs: 2.1 km at 20° N of E, 3.4 km at 35° W of N, 1.8 km S, 2.6 km at 10° S of W. Where does the rider end up?

- Graphical methods give physical understanding ...
- ... but scale drawings have limited precision,
- and repeated triangle calculations become cumbersome.
- **Components** provide a systematic method for any number of vectors.

Next time

Components turn a geometrical vector problem into an algebraic one.

Contact 2: Calculate

How do we calculate with vectors accurately?

Graphical methods

- Show what is happening
- Good for sketches and checks
- Limited accuracy; awkward for many vectors

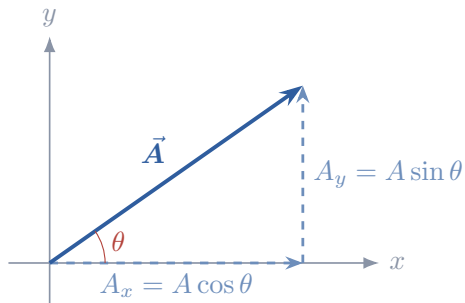
Components

- Turn vectors into ordinary numbers
- Any number of vectors, any accuracy
- Work the same way in 3D

Physical meaning

Always sketch first, then predict, then calculate with components, then use the sketch to check the answer.

What is a component?



θ measured from $+x$ toward $+y$

A_x tells us **how much of $\vec{A}$ lies along the x -direction**; A_y , how much lies along the y -direction.

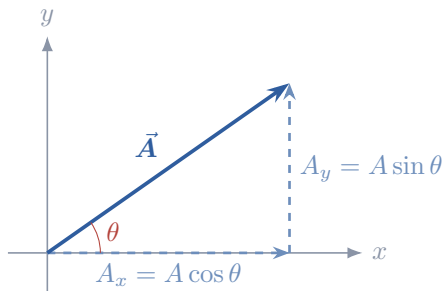
Scalar components, not vectors

A_x and A_y are **numbers**: positive, negative or zero.

The **component vectors** are $A_x \hat{i}$ and $A_y \hat{j}$ ($\hat{i}, \hat{j}$ point along $+x, +y$; more shortly), so

$$\vec{A} = A_x \hat{i} + A_y \hat{j}.$$

Components of a vector



θ measured from $+x$ toward $+y$

Rule: components (2D)

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

These apply when θ is measured from the $+x$ -axis toward the $+y$ -axis.

Angle measured from another axis?

Either convert it to the standard θ , or decide whether the component is *adjacent* or *opposite* to the given angle and assign the signs from the sketch (Toolkit 3).

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch → predict signs → calculate

Right or left? Up or down? So what signs do you expect?

Question

A force of 12.0 N acts at $\theta = 150^\circ$ (counter-clockwise from $+x$). Find F_x and F_y .

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**

Right or left? Up or down? So what signs do you expect?

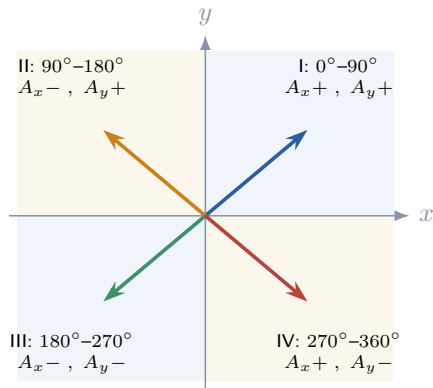
Quadrant II, so $F_x < 0$, $F_y > 0$.

$$F_x = 12.0 \cos 150^\circ = -10.4 \text{ N}$$

$$F_y = 12.0 \sin 150^\circ = +6.00 \text{ N}$$

Evaluate: $\sqrt{10.4^2 + 6.00^2} = 12.0 \checkmark$

Predict the signs first



Before using equations, ask: does the vector point **right or left**? Does it point **up or down**?

Sign rule

quadrant	I	II	III	IV
A_x	> 0	< 0	< 0	> 0
A_y	> 0	> 0	< 0	< 0

Permanent rule

Sketch → **predict signs** → **calculate**
The calculator never replaces the sketch.

TOOLKIT 3 – Components

Choose axes → **sketch** → **predict signs** → **resolve**

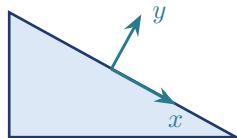
Choose axes. Mutually perpendicular, not necessarily horizontal and vertical: x = east, y = north for displacements; x along a slope for inclined planes.

Sketch and predict signs. Right or left gives the sign of A_x ; up or down gives the sign of A_y .

Resolve. $A_x = A \cos \theta$, $A_y = A \sin \theta$ when θ is measured from $+x$. In general:

Relative to the stated angle:
adjacent → **cosine**; **opposite** → **sine**.
The signs come from the sketch.

Components and component vectors. A_x , A_y are scalars (positive, negative or zero); $A_x \hat{i}$, $A_y \hat{j}$ are vectors, and $\vec{A} = A_x \hat{i} + A_y \hat{j}$.



axes tilted with a slope

Your turn: angles measured from other axes

Your turn

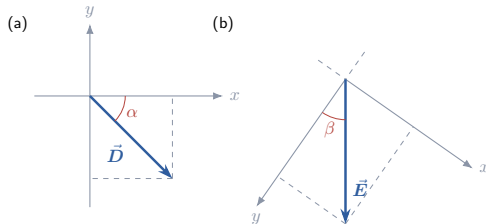
Find the x - and y -components of $\vec{D}$ and $\vec{E}$.

(a) $|\vec{D}| = 3.00 \text{ m}$, $\alpha = 45^\circ$ below $+x$.

(b) $|\vec{E}| = 4.50 \text{ m}$, $\beta = 37.0^\circ$ from the $+y$ axis, with tilted axes.

Sketch $\rightarrow$ **predict signs** $\rightarrow$ **calculate**.

Young, Example 1.6.



(a) Measured counter-clockwise from $+x$,
 $\theta = -45^\circ$:

$$D_x = 3.00 \cos(-45^\circ) = +2.12 \text{ m}$$

$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

(a) Measured counter-clockwise from $+x$,
 $\theta = -45^\circ$:

$$D_x = 3.00 \cos(-45^\circ) = +2.12 \text{ m}$$

$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

(b) β is measured from $+y$, so E_y is *adjacent*
and E_x is *opposite*:

$$E_x = 4.50 \sin 37.0^\circ = +2.71 \text{ m}$$

$$E_y = 4.50 \cos 37.0^\circ = +3.59 \text{ m}$$

Both positive, since $\vec{E}$ lies between the $+x$ and $+y$ axes of the tilted system.

(a) Measured counter-clockwise from $+x$,
 $\theta = -45^\circ$:

$$D_x = 3.00 \cos(-45^\circ) = +2.12 \text{ m}$$

$$D_y = 3.00 \sin(-45^\circ) = -2.12 \text{ m}$$

(b) β is measured from $+y$, so E_y is *adjacent*
and E_x is *opposite*:

$$E_x = 4.50 \sin 37.0^\circ = +2.71 \text{ m}$$

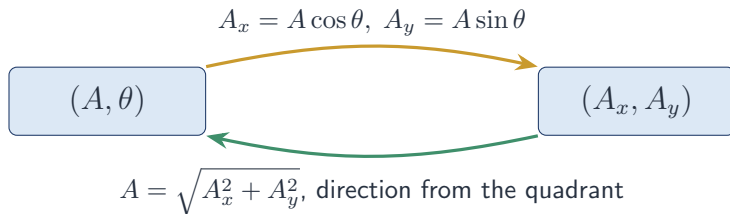
$$E_y = 4.50 \cos 37.0^\circ = +3.59 \text{ m}$$

Both positive, since $\vec{E}$ lies between the $+x$ and $+y$ axes of the tilted system.

Physical meaning

The formulas $A \cos \theta$, $A \sin \theta$ only hold for θ measured from $+x$. Otherwise think “adjacent or opposite?”

Two equivalent descriptions of one vector



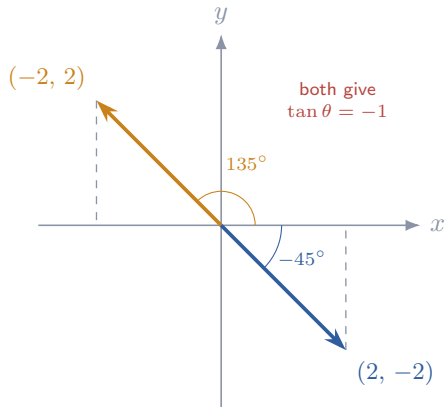
Key idea

Magnitude/direction and components are two equivalent ways of describing the same vector. Going from components back to magnitude and direction is Toolkit 4.

Direction from components: a common error

Predict first

$\vec{P} = (-2, 2)$ and $\vec{Q} = (2, -2)$. Both have $A_y/A_x = -1$, and a calculator gives $\tan^{-1}(-1) = -45^\circ$ for both. Same direction?



TOOLKIT 4 – Reconstructing a vector

Magnitude: $A = \sqrt{A_x^2 + A_y^2}$.

Direction (do not trust $\tan^{-1}$ alone):

1. Identify the quadrant from the signs of A_x , A_y .
2. Reference angle (the acute angle to the x -axis): $\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|$.
3. Place it in the quadrant:
I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$;
III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.
4. Check against the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions.

Quadrant first, calculation second.

$\vec{P} = (-2, 2)$: left and up, quadrant II.

$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135^\circ}$$

TOOLKIT 4 – Reconstructing a vector

Magnitude: $A = \sqrt{A_x^2 + A_y^2}$.

Direction (do not trust $\tan^{-1}$ alone):

1. Identify the quadrant from the signs of A_x , A_y .
2. Reference angle (the acute angle to the x -axis): $\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|$.
3. Place it in the quadrant:
I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$;
III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.
4. Check against the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions.
Quadrant first, calculation second.

$\vec{P} = (-2, 2)$: left and up, quadrant II.

$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135^\circ}$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.

$$\theta_{\text{ref}} = 45^\circ$$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315^\circ} (= -45^\circ)$$

TOOLKIT 4 – Reconstructing a vector

Magnitude: $A = \sqrt{A_x^2 + A_y^2}$.

Direction (do not trust $\tan^{-1}$ alone):

1. Identify the quadrant from the signs of A_x , A_y .
2. Reference angle (the acute angle to the x -axis): $\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|$.
3. Place it in the quadrant:
I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$;
III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.
4. Check against the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions.

Quadrant first, calculation second.

$\vec{P} = (-2, 2)$: left and up, quadrant II.

$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135^\circ}$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.

$$\theta_{\text{ref}} = 45^\circ$$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315^\circ} (= -45^\circ)$$

Your turn

$A_x = -3.0 \text{ m}$, $A_y = -4.0 \text{ m}$. Find A and θ .

TOOLKIT 4 – Reconstructing a vector

Magnitude: $A = \sqrt{A_x^2 + A_y^2}$.

Direction (do not trust $\tan^{-1}$ alone):

1. Identify the quadrant from the signs of A_x , A_y .
2. Reference angle (the acute angle to the x -axis): $\theta_{\text{ref}} = \tan^{-1} |A_y/A_x|$.
3. Place it in the quadrant:
I: θ_{ref} ; II: $180^\circ - \theta_{\text{ref}}$;
III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.
4. Check against the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions.
Quadrant first, calculation second.

$\vec{P} = (-2, 2)$: left and up, quadrant II.

$$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135^\circ}$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.

$$\theta_{\text{ref}} = 45^\circ$$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315^\circ} (= -45^\circ)$$

Your turn

$A_x = -3.0$ m, $A_y = -4.0$ m. Find A and θ .

$A = 5.0$ m; quadrant III;

$$\theta_{\text{ref}} = \tan^{-1}(4/3) = 53.1^\circ;$$

$$\theta = 180^\circ + 53.1^\circ = 233.1^\circ \checkmark$$

Question

If $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \cdots$, how are the components of $\vec{R}$ related to theirs?

Adding vectors with components

Question

If $\vec{R} = \vec{A} + \vec{B} + \vec{C} + \dots$, how are the components of $\vec{R}$ related to theirs?

Rule: adding vectors by components

$$R_x = \sum A_x, \quad R_y = \sum A_y \quad (\text{in 3D also } R_z = \sum A_z)$$

✂ TOOLKIT 5 – Adding vectors by components: RESOLVE → COMBINE → RECONSTRUCT

RESOLVE: find the components of every vector (a table is often useful). **Sketch → predict signs → calculate.**

COMBINE: $R_x = \sum A_x$, $R_y = \sum A_y$, and in 3D $R_z = \sum A_z$.

RECONSTRUCT: find the magnitude and direction of $\vec{R}$ (Toolkit 4).

This is the principal computational technique of the week.

Lecturer example: three displacements (Young Ex. 1.7)

Question

A contestant in a radio race makes three displacements:

$\vec{A}$: 72.4 m, 32.0° **east of north**

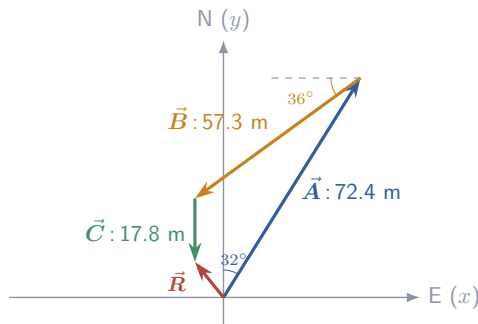
$\vec{B}$: 57.3 m, 36.0° **south of west**

$\vec{C}$: 17.8 m **south**

Find the resultant $\vec{R}$.

Predict first

From the sketch: roughly how far, and in which quadrant?



Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

	magnitude	θ	x -component	y -component
$\vec{A}$	72.4 m	58.0°	38.37 m	61.40 m
$\vec{B}$	57.3 m	216.0°	-46.36 m	-33.68 m
$\vec{C}$	17.8 m	270.0°	0.00 m	-17.80 m
$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

Identify: target $\vec{R}$; known three displacements. **Set up:** x east, y north; convert each direction to θ from $+x$: $\theta_A = 90^\circ - 32.0^\circ = 58.0^\circ$, $\theta_B = 180^\circ + 36.0^\circ = 216.0^\circ$, $\theta_C = 270^\circ$.

	magnitude	θ	x -component	y -component
$\vec{A}$	72.4 m	58.0°	38.37 m	61.40 m
$\vec{B}$	57.3 m	216.0°	-46.36 m	-33.68 m
$\vec{C}$	17.8 m	270.0°	0.00 m	-17.80 m
$\vec{R}$			$R_x = -7.99 \text{ m}$	$R_y = 9.92 \text{ m}$

Execute (resolve, combine, reconstruct): $R = \sqrt{(-7.99)^2 + 9.92^2} = 12.7 \text{ m}$; left and up, quadrant II; $\theta_{\text{ref}} = \tan^{-1}\left(\frac{9.92}{7.99}\right) = 51.1^\circ$, so $\theta = 180^\circ - 51.1^\circ = 129^\circ$, i.e. 12.7 m at 39° west of north. **Evaluate:** units m; quadrant II, matching the sketch. ✓

Stating the direction in words

A calculation may give $\theta = 129^\circ$, but the physical answer is often clearer as a compass direction.

Measure from the nearest cardinal axis:

$$\text{north} = 90^\circ, \quad 129^\circ - 90^\circ = 39^\circ$$

so $\theta = 129^\circ$ is 39° **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1^\circ$ and $\theta = 96.2^\circ$ as compass directions.

Stating the direction in words

A calculation may give $\theta = 129^\circ$, but the physical answer is often clearer as a compass direction.

Measure from the nearest cardinal axis:

$$\text{north} = 90^\circ, \quad 129^\circ - 90^\circ = 39^\circ$$

so $\theta = 129^\circ$ is 39° **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1^\circ$ and $\theta = 96.2^\circ$ as compass directions.

$233.1^\circ - 180^\circ = 53.1^\circ$ south of west;
 $96.2^\circ - 90^\circ = 6.2^\circ$ west of north.

Your turn

A hiker walks 4.00 km at 30.0° north of east, then 6.00 km at 40.0° west of north, then 3.00 km south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{R}$ be in?

Your turn

A hiker walks 4.00 km at 30.0° north of east, then 6.00 km at 40.0° west of north, then 3.00 km south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{R}$ be in?

θ : 30.0° , 130.0° , 270°

$$R_x = 3.46 - 3.86 + 0 = -0.39 \text{ km}$$

$$R_y = 2.00 + 4.60 - 3.00 = 3.60 \text{ km}$$

Your turn

A hiker walks 4.00 km at 30.0° north of east, then 6.00 km at 40.0° west of north, then 3.00 km south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{R}$ be in?

θ : 30.0° , 130.0° , 270°

$$R_x = 3.46 - 3.86 + 0 = -0.39 \text{ km}$$

$$R_y = 2.00 + 4.60 - 3.00 = 3.60 \text{ km}$$

$$R = 3.62 \text{ km}$$

left and up: quadrant II

$$\theta_{\text{ref}} = \tan^{-1}(3.60/0.39) = 83.8^\circ$$

$\theta = 180^\circ - 83.8^\circ = 96.2^\circ$, i.e. 6.2° west of north.

Unit vectors

Unit vectors $\hat{i}, \hat{j}, \hat{k}$

Magnitude 1, no units, pointing along $+x$, $+y$, $+z$.

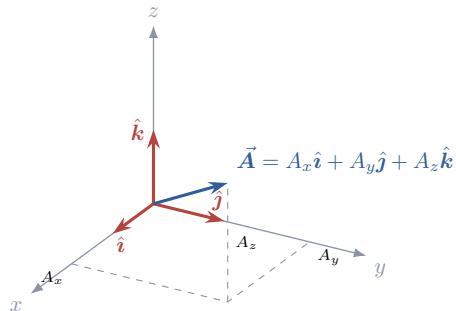
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

A compact way to write components *together with their directions*.

Rule: adding in unit-vector form

Collect like terms:

$$\begin{aligned}\vec{A} + \vec{B} &= (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} \\ &\quad + (A_z + B_z)\hat{k}\end{aligned}$$



Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) = (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m}$$

$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) = (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m}$$

$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Your turn

$\vec{A} = (4\hat{i} - 5\hat{j} + 7\hat{k})$ m, $\vec{B} = (\hat{i} - 3\hat{j} + 5\hat{k})$ m, $\vec{C} = (\hat{i} + \hat{j} - 2\hat{k})$ m. Find $\vec{A} + \vec{B} + \vec{C}$ and its magnitude.

Question

$\vec{D} = (6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k})$ m and $\vec{E} = (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k})$ m. Find $\vec{F} = 2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{i} + 3.00\hat{j} - 1.00\hat{k}) - (4.00\hat{i} - 5.00\hat{j} + 8.00\hat{k}) = (8.00\hat{i} + 11.00\hat{j} - 10.00\hat{k}) \text{ m}$$

$$F = \sqrt{(8.00 \text{ m})^2 + (11.00 \text{ m})^2 + (-10.00 \text{ m})^2} = 16.9 \text{ m}$$

Your turn

$\vec{A} = (4\hat{i} - 5\hat{j} + 7\hat{k})$ m, $\vec{B} = (\hat{i} - 3\hat{j} + 5\hat{k})$ m, $\vec{C} = (\hat{i} + \hat{j} - 2\hat{k})$ m. Find $\vec{A} + \vec{B} + \vec{C}$ and its magnitude. $(6\hat{i} - 7\hat{j} + 10\hat{k})$ m, $|\cdot| = \sqrt{185} = 13.6 \text{ m}$

Contact 3: Extend

How do vector products describe physical quantities?

The 3D Cartesian system

Rule: right-handed axes

Curl the fingers of the right hand from $+x$ to $+y$; the thumb points along $+z$.

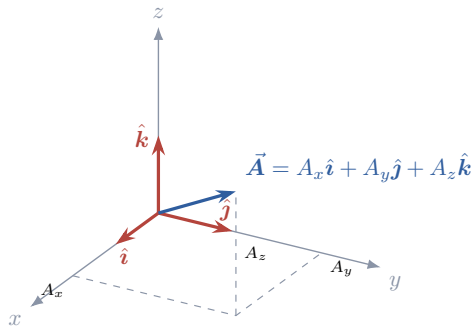
Rule: components and magnitude in 3D

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Physical meaning

The mutually perpendicular unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ provide three independent directions, so a vector is uniquely specified by its x -, y - and z -components.



Can vectors be multiplied?

Question

We can add and subtract vectors. Can we also multiply one vector by another?

Can vectors be multiplied?

Question

We can add and subtract vectors. Can we also multiply one vector by another?

There are two important vector products:

Dot product

vector $\cdot$ vector $\rightarrow$ **scalar**

Cross product

vector $\times$ vector $\rightarrow$ **vector**

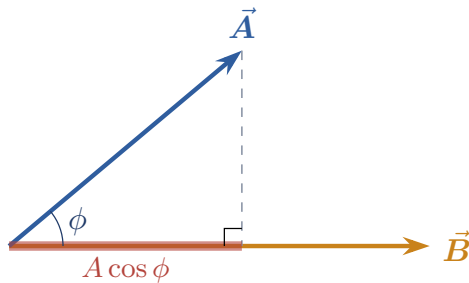
Physical meaning

We first understand what each product means geometrically, then learn how to calculate it.

The dot (scalar) product

Question

How much of one vector lies along the direction of another?



The part of $\vec{A}$ that lies along $\vec{B}$ is the **projection**

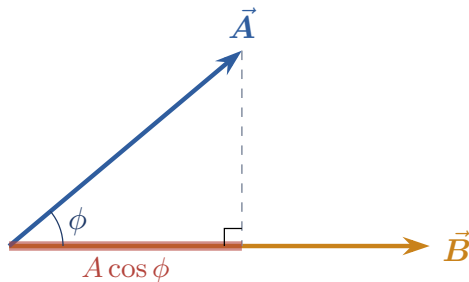
$$A \cos \phi$$

where ϕ is the angle between the vectors when placed tail-to-tail, conventionally taken in the range $0^\circ \leq \phi \leq 180^\circ$.

The dot (scalar) product

Question

How much of one vector lies along the direction of another?



The part of $\vec{A}$ that lies along $\vec{B}$ is the **projection**

$$A \cos \phi$$

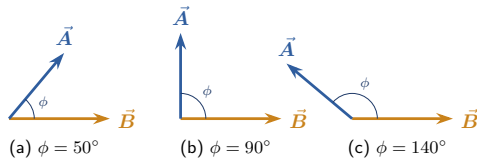
where ϕ is the angle between the vectors when placed tail-to-tail, conventionally taken in the range $0^\circ \leq \phi \leq 180^\circ$.

Definition: dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi = (A \cos \phi) B$$

vector $\cdot$ vector $\rightarrow$ **scalar**. Units multiply:
 $\text{m} \cdot \text{m} = \text{m}^2$.

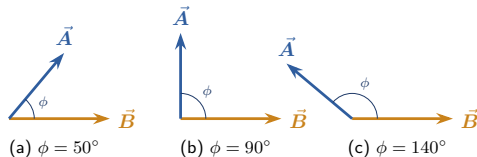
Predict the sign of $\vec{A} \cdot \vec{B}$



Predict first

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

Predict the sign of $\vec{A} \cdot \vec{B}$



Predict first

For each pair: is $\vec{A} \cdot \vec{B}$ positive, zero, or negative?

Rule: sign of the dot product

$$\phi < 90^\circ \Rightarrow \vec{A} \cdot \vec{B} > 0; \quad \phi = 90^\circ \Rightarrow \vec{A} \cdot \vec{B} = 0; \quad \phi > 90^\circ \Rightarrow \vec{A} \cdot \vec{B} < 0$$

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Rule: unit-vector dot products

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

Question

What are $\hat{i} \cdot \hat{i}$ and $\hat{i} \cdot \hat{j}$?

Rule: unit-vector dot products

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

Expanding $(A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) \cdot (B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$, only the like terms survive:

Rule: dot product from components

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Physics preview: Work

$$W = \vec{F} \cdot \vec{s}$$

A constant force $\vec{F}$ acting through a displacement $\vec{s}$. Work is studied fully later in the course.

Question

$|\vec{A}| = 7.50 \text{ m}$ at 53.0° , $|\vec{B}| = 5.00 \text{ m}$ at 130.0° . Find $\vec{A} \cdot \vec{B}$.

Predict first

Is ϕ less than or more than 90° ? So what sign?

Lecturer example: dot product from a figure

Question

$|\vec{A}| = 7.50 \text{ m}$ at 53.0° , $|\vec{B}| = 5.00 \text{ m}$ at 130.0° . Find $\vec{A} \cdot \vec{B}$.

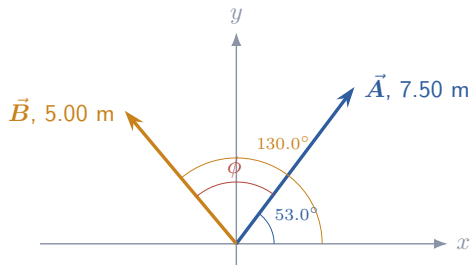
Predict first

Is ϕ less than or more than 90° ? So what sign?

$$\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ$$

$$\vec{A} \cdot \vec{B} = (7.50 \text{ m})(5.00 \text{ m}) \cos 77.0^\circ = 8.44 \text{ m}^2$$

Evaluate: positive since $\phi < 90^\circ$; units are m^2 , not m.



after Young, Example 1.9

TOOLKIT 6 – Dot product

Two forms, one quantity:

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Predict the sign first:

acute angle $\rightarrow$ positive; perpendicular $\rightarrow$ zero;
obtuse angle $\rightarrow$ negative.

CHECK: the calculated sign must agree with the geometry: positive $\rightarrow \phi < 90^\circ$,
zero $\rightarrow \phi = 90^\circ$, negative $\rightarrow \phi > 90^\circ$.

Angle between two vectors:

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB}$$

with $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$ and
 $B = \sqrt{B_x^2 + B_y^2 + B_z^2}$.

Physics preview: work, $W = \vec{F} \cdot \vec{s}$.

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn

Find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn

Find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}, \quad \vec{B} = -5\hat{i} + \hat{j} - \hat{k}$$

Your turn

Find $\vec{A} \cdot \vec{B}$ and the angle ϕ between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

$$A = \sqrt{1 + 81 + 4} = \sqrt{86} = 9.27, \quad B = \sqrt{25 + 1 + 1} = \sqrt{27} = 5.20$$

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{2}{(9.27)(5.20)} = 0.0415 \Rightarrow \phi = 87.6^\circ$$

Evaluate: small positive dot product $\Rightarrow$ angle just under 90° . ✓

Dot product and cross product compared

Dot product

$$\vec{A} \cdot \vec{B} = AB \cos \phi$$

- Result: a **scalar**
- Zero when the vectors are perpendicular
- Maximum magnitude when the vectors are parallel or antiparallel

Cross product

$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

- Result: a **vector**
- Zero when the vectors are parallel or antiparallel
- Maximum magnitude when the vectors are perpendicular

The angle ϕ in both products

ϕ is the **angle between the vectors when placed tail-to-tail, conventionally taken in the range $0^\circ \leq \phi \leq 180^\circ$** . It is not an internal angle of a head-to-tail triangle, such as the 120° used in the cyclist problem.

The cross (vector) product: geometry first

Question

How can two vectors produce a *third* vector?

The cross (vector) product: geometry first

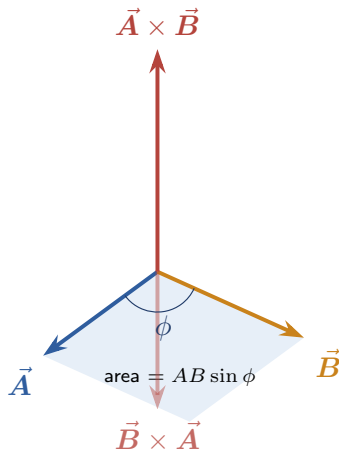
Question

How can two vectors produce a *third* vector?

Definition: cross product

- Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$ (area of the parallelogram)
- Direction: perpendicular to the plane of $\vec{A}$ and $\vec{B}$, given by the **right-hand rule**

The right-hand rule is on the next slide.



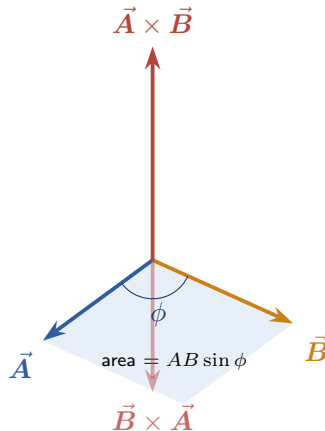
Right-hand rule

For $\vec{A} \times \vec{B}$:

1. Place $\vec{A}$ and $\vec{B}$ tail to tail.
2. Point the fingers of the right hand along $\vec{A}$.
3. Curl the fingers toward $\vec{B}$ through the smaller angle.
4. The thumb gives the direction of $\vec{A} \times \vec{B}$.

Caution

The order matters: $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$.
Swapping the vectors reverses the direction.



Properties of the cross product

Predict first

What is $\vec{A} \times \vec{A}$? What happens if you swap the order?

Properties of the cross product

Predict first

What is $\vec{A} \times \vec{A}$? What happens if you swap the order?

Rules: properties of the cross product

$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (the order matters)

Parallel or antiparallel vectors:

$$\sin \phi = 0 \Rightarrow \vec{A} \times \vec{B} = \vec{0}$$

$\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$

Properties of the cross product

Predict first

What is $\vec{A} \times \vec{A}$? What happens if you swap the order?

Rules: properties of the cross product

$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (the order matters)

Parallel or antiparallel vectors:

$$\sin \phi = 0 \Rightarrow \vec{A} \times \vec{B} = \vec{0}$$

$\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$

Rule: unit-vector cross products

Cyclic order $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives +: $\hat{i} \times \hat{j} = \hat{k}$,
 $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$.

Reversed order gives -: $\hat{j} \times \hat{i} = -\hat{k}$, $\hat{k} \times \hat{j} = -\hat{i}$,
 $\hat{i} \times \hat{k} = -\hat{j}$.

Also $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$.

$\times$	$\hat{i}$	$\hat{j}$	$\hat{k}$
$\hat{i}$	$\vec{0}$	$\hat{k}$	$-\hat{j}$
$\hat{j}$	$-\hat{k}$	$\vec{0}$	$\hat{i}$
$\hat{k}$	$\hat{j}$	$-\hat{i}$	$\vec{0}$

row $\times$ column

Rule: cross product from components

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

Memory aid: the determinant

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

The $\hat{j}$ term takes a minus sign when the determinant is expanded.

Physics preview: Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

A force $\vec{F}$ applied at position $\vec{r}$ from a pivot. Torque is studied fully later in the course.

✂ TOOLKIT 7 – Cross product: technical summary

Magnitude: $|\vec{A} \times \vec{B}| = AB \sin \phi$.

Direction: right-hand rule; $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$.

Predict before calculating:

parallel or antiparallel $\rightarrow \vec{A} \times \vec{B} = \vec{0}$

($\sin 0^\circ = \sin 180^\circ = 0$);

perpendicular $\rightarrow$ maximum magnitude

($\sin 90^\circ = 1$).

Unit vectors: cyclic $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives +;
reversed order gives -; $\hat{i} \times \hat{i} = \vec{0}$.

Components:

$$(A_y B_z - A_z B_y) \hat{i}$$

$$+ (A_z B_x - A_x B_z) \hat{j}$$

$$+ (A_x B_y - A_y B_x) \hat{k}$$

Perpendicularity check:

$$(\vec{A} \times \vec{B}) \cdot \vec{A} = 0, \quad (\vec{A} \times \vec{B}) \cdot \vec{B} = 0$$

Physics preview: torque, $\vec{\tau} = \vec{r} \times \vec{F}$.

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Question

$\vec{A}$ has magnitude 6 units along $+x$. $\vec{B}$ has magnitude 4 units in the xy -plane at 30° to $+x$. Find $\vec{A} \times \vec{B}$.

Predict first

Before calculating: which way will $\vec{A} \times \vec{B}$ point?

Magnitude:

$$AB \sin \phi = (6)(4) \sin 30^\circ = 12$$

Direction: right-hand rule from $+x$ towards $\vec{B}$ gives $+z$.

$$\vec{A} \times \vec{B} = 12 \hat{k}$$

Check (components): $\vec{A} = 6\hat{i}$,

$$\vec{B} = 3.46\hat{i} + 2\hat{j}$$

$$\Rightarrow (6)(2) - (0)(3.46) = 12 \text{ along } \hat{k}. \checkmark$$

Your turn: the same vector pair

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

$$(\vec{A} \times \vec{B})_x = A_y B_z - A_z B_y = (9)(-1) - (2)(1) = -11$$

$$(\vec{A} \times \vec{B})_y = A_z B_x - A_x B_z = (2)(-5) - (1)(-1) = -9$$

$$(\vec{A} \times \vec{B})_z = A_x B_y - A_y B_x = (1)(1) - (9)(-5) = 46$$

$$\vec{A} \times \vec{B} = -11\hat{i} - 9\hat{j} + 46\hat{k}$$

Your turn

$\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$, $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. Find $\vec{A} \times \vec{B}$, then check it (Toolkit 7).

$$(\vec{A} \times \vec{B})_x = A_y B_z - A_z B_y = (9)(-1) - (2)(1) = -11$$

$$(\vec{A} \times \vec{B})_y = A_z B_x - A_x B_z = (2)(-5) - (1)(-1) = -9$$

$$(\vec{A} \times \vec{B})_z = A_x B_y - A_y B_x = (1)(1) - (9)(-5) = 46$$

$$\vec{A} \times \vec{B} = -11\hat{i} - 9\hat{j} + 46\hat{k}$$

Perpendicularity check (Toolkit 7):

$$(\vec{A} \times \vec{B}) \cdot \vec{A} = -11 - 81 + 92 = 0 \quad \checkmark \qquad (\vec{A} \times \vec{B}) \cdot \vec{B} = 55 - 9 - 46 = 0 \quad \checkmark$$

Integrated problem: pulling a crate

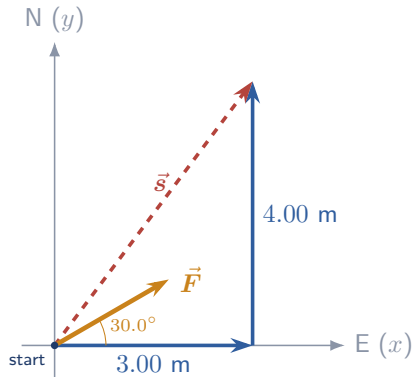
Integrated problem

A worker pulls a crate across a level floor with a rope. The rope force is 50.0 N at 30.0° north of east. The crate moves 3.00 m east, then 4.00 m north.

- (a) Find the crate's displacement $\vec{s}$.
- (b) Find the work done by the rope force, $W = \vec{F} \cdot \vec{s}$.
- (c) The same force acts at $\vec{r} = (0.800\hat{i} + 0.600\hat{j})\text{ m}$ from a pivot. Find the torque $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Sketch first. Is W positive, zero or negative?
Does $\vec{\tau}$ point along $+z$ or $-z$?



top view; $F = 50.0\text{ N}$; z points out of the page

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (Toolkit 5):

$$\vec{s} = (3.00\hat{i} + 4.00\hat{j}) \text{ m}$$

$$s = 5.00 \text{ m at } 53.1^\circ \text{ north of east}$$

Resolve the force (Toolkit 3):

$$F_x = 50.0 \cos 30.0^\circ = 43.3 \text{ N}$$

$$F_y = 50.0 \sin 30.0^\circ = 25.0 \text{ N}$$

(b) Dot product (Toolkit 6):

$$W = (43.3)(3.00) + (25.0)(4.00) = 230 \text{ J}$$

(c) Cross product (Toolkit 7); only the $\hat{k}$ term survives:

$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Identify: targets $\vec{s}$, W and $\vec{\tau}$. **Set up:** x east, y north, z up; $\vec{F}$ at $\theta = 30.0^\circ$ (Toolkit 1). Predict: $\phi < 90^\circ$, so $W > 0$.

Execute.

(a) Add the legs (Toolkit 5):

$$\vec{s} = (3.00\hat{i} + 4.00\hat{j}) \text{ m}$$

$$s = 5.00 \text{ m at } 53.1^\circ \text{ north of east}$$

Resolve the force (Toolkit 3):

$$F_x = 50.0 \cos 30.0^\circ = 43.3 \text{ N}$$

$$F_y = 50.0 \sin 30.0^\circ = 25.0 \text{ N}$$

(b) Dot product (Toolkit 6):

$$W = (43.3)(3.00) + (25.0)(4.00) = 230 \text{ J}$$

(c) Cross product (Toolkit 7); only the $\hat{k}$ term survives:

$$\tau_z = r_x F_y - r_y F_x = (0.800)(25.0) - (0.600)(43.3)$$

$$\vec{\tau} = -5.98 \hat{k} \text{ N m}$$

Evaluate (verification):

$$Fs \cos \phi = (50.0)(5.00) \cos 23.1^\circ = 230 \text{ J } \checkmark$$

$$rF \sin \phi = (1.00)(50.0) \sin 6.9^\circ = 5.98 \text{ N m } \checkmark$$

Right-hand rule from $\vec{r}$ (36.9°) to $\vec{F}$ (30.0°):

clockwise seen from above, so $-z$ $\checkmark$

$W > 0$, as predicted $\checkmark$

Your turn: an integrated problem

Your turn

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

Your turn: an integrated problem

Your turn

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

(a) $\vec{s} = (-2.00\hat{i} + 2.00\hat{j})$ m, $s = 2.83$ m at 45° north of west

(b) $F_x = 20.0$ N, $F_y = 34.6$ N:

$$W = (20.0)(-2.00) + (34.6)(2.00) = 29.3 \text{ J}$$

Your turn: an integrated problem

Your turn

A rope force of 40.0 N acts at 60.0° north of east. The object moves 2.00 m north, then 2.00 m west.

(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{i}$ m from a pivot; find $\vec{\tau}$.

Verify (b) with $Fs \cos \phi$ and (c) with $rF \sin \phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90° ?

(a) $\vec{s} = (-2.00\hat{i} + 2.00\hat{j})$ m, $s = 2.83$ m at 45° north of west

(b) $F_x = 20.0$ N, $F_y = 34.6$ N:
 $W = (20.0)(-2.00) + (34.6)(2.00) = 29.3$ J

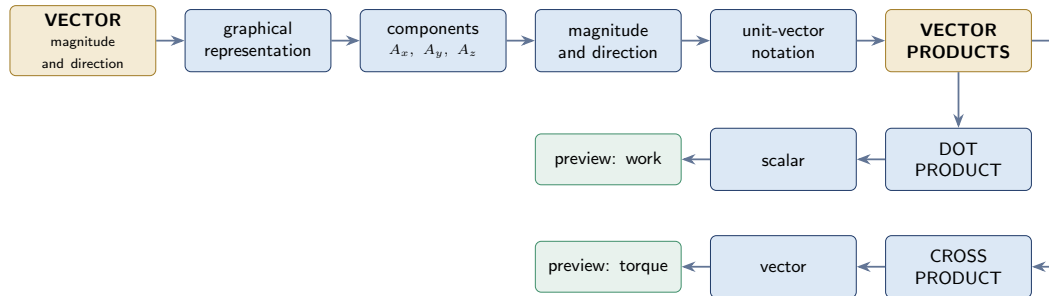
Check: $\phi = 135^\circ - 60^\circ = 75^\circ$;

$(40.0)(2.83) \cos 75^\circ = 29.3$ J ✓

(c) $\tau_z = r_x F_y - r_y F_x = (0.500)(34.6) - 0 = 17.3$, so $\vec{\tau} = 17.3 \hat{k}$ N m

Check: $(0.500)(40.0) \sin 60^\circ = 17.3$ N m;
counter-clockwise from above ✓

Week 1 concept map





MASTER TOOLKIT – For a general vector problem

1. **IDENTIFY** What is being asked?
2. **SKETCH** Draw the physical situation.
3. **CHOOSE AXES** Select useful perpendicular coordinates.
4. **PREDICT** Estimate signs, quadrant, size and direction.
5. **RESOLVE** Convert vectors into components if necessary.
6. **COMBINE** Perform the required vector operation.
7. **RECONSTRUCT** Find magnitude and direction if required.
8. **EVALUATE** Check units, signs, quadrant, magnitude, direction and physical reasonableness.

If the result disagrees with the sketch or prediction, investigate before accepting the answer.

Inside the course framework

IDENTIFY $\leftrightarrow$ step 1

SET UP $\leftrightarrow$ steps 2–4

EXECUTE $\leftrightarrow$ steps 5–7

EVALUATE $\leftrightarrow$ step 8

Four habits for Week 1

✓ Habit 1

Sketch before calculating.

✓ Habit 2

Predict signs before pressing the calculator.

✓ Habit 3

Determine the quadrant before accepting an angle.

✓ Habit 4

Check whether the final answer makes physical sense.

THE SEVEN TOOLKITS

- | | | |
|----------|--------------------------------|--|
| 1 | Directions and angles | directional language, standard angle, quadrant, reference angle, compass directions |
| 2 | Vector triangles | head to tail, internal angle, Pythagoras or sine and cosine rules, $ A - B \leq R \leq A + B$ |
| 3 | Components | choose axes $\rightarrow$ sketch $\rightarrow$ predict signs $\rightarrow$ resolve |
| 4 | Reconstructing a vector | magnitude; direction by quadrant, reference angle, placement, check |
| 5 | Adding by components | RESOLVE $\rightarrow$ COMBINE $\rightarrow$ RECONSTRUCT |
| 6 | Dot product | two forms, sign check, angle between vectors; preview: work |
| 7 | Cross product | magnitude, right-hand rule, unit vectors, components, perpendicularity check; preview: torque |

Extension: direction angles in 3D are in the supplementary notes.

Checklist: can I ...

- ☐ tell a scalar from a vector?
- ☐ add and subtract vectors graphically?
- ☐ use the sine and cosine rules on a vector triangle?
- ☐ resolve a vector with the correct signs?
- ☐ get the correct quadrant for θ ?
- ☐ add several vectors by components?
- ☐ write a vector with $\hat{i}, \hat{j}, \hat{k}$?
- ☐ use IDENTIFY $\rightarrow$ SET UP $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE?
- ☐ calculate a dot product?
- ☐ determine whether a dot product is positive, zero or negative?
- ☐ use the dot product to find the angle between two vectors?
- ☐ calculate a cross product?
- ☐ use the right-hand rule?
- ☐ explain when a cross product is zero?

Next

Problem sheet 1 on Sakai: extra component, resultant, dot and cross problems, plus challenge problems. Read Young §1.7–1.10. Supplementary notes cover linear independence and direction angles in 3D (extension).